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CODEX OF THE TABLE

*Nutrition with both columns of the table: what to take,
what not to exceed, and where every figure comes from*

VILLUMINATIONS

Fine edition, 2026

Original work · First edition, 2026
villuminations.com

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CHAPTER 1

WHAT THIS BOOK CLAIMS

This book has no method. There is no VILLUMINATIONS diet, there are no forbidden foods and there is no list of supplements at the end. I say it in the first line because it is the opposite of what a nutrition book usually offers, and because the rest only makes sense with this in front of you.

THE CLAIM

There is a body of human nutrition data that is both first rate and freely accessible, and almost nobody opens it. It is in the Dietary Reference Intakes, in the official dietary guidelines, in the food composition database of the U.S. Department of Agriculture, and in the field manuals of an army that feeds and trains a million people and publishes its doctrine in full. All of it is free, it is online, and no copyright protects it.

What copyright does protect, and what does cost money, are the interpretations: the diet books, the paid plans, the influencers. This book goes the other way. It goes to the source, takes out the whole table—not the pretty half—and explains how to read it.

WHY THE MISSING HALF OF THE TABLE MATTERS SO MUCH

Any nutrition leaflet prints a nutrient's recommended amount. Very few print its tolerable upper limit, which is the other column of the same official document. The result is a public that knows how much magnesium it “needs” and has no idea at what point the problem starts, at a moment in history when supplementing is as easy as ordering a parcel.

This book prints both columns for all nineteen nutrients, and devotes a whole chapter to ranking them by how narrow the margin between the two is. That ranking is a calculation, not an opinion, and it completely changes your idea of what can be taken blind.



WHAT THIS BOOK DOES NOT CLAIM

- It is not an individual prescription. Reference values are population values: they describe what covers the needs of almost all healthy adults, not yours.
- It does not account for medical conditions, pregnancy, breastfeeding, childhood, advanced age or medication. Any of those words changes the figures, and in some cases reverses them.
- It does not replace a health professional. A blood test and an appointment are worth more than this entire book, and this book is written so that you arrive at that appointment knowing what to ask.

- It promises no body composition results. No figure in here makes anyone leaner; what they do is stop you going short of something while you do what actually works.

A WARNING THAT TRAVELS WITH THE DATA

If you take medication —anticoagulants, thyroid drugs, antihypertensives, metformin, lithium, among others— there are nutrients in this table that interact with it in clinically relevant ways. Vitamin K and anticoagulants are the textbook case, but not the only one. Do not change an intake or start a supplement without telling whoever prescribes for you. This is not a polite formula: it is the most useful piece of advice in the book.



WHERE EVERY FIGURE COMES FROM

From eight public sources, all named in the final chapter with an indication of where to consult them. No figure in this book is invented, rounded for convenience, or copied from another popular-science book.

And there is something else, which is what makes this book a system rather than a well-laid-out leaflet: the tables check themselves. The programme that typesets these pages first runs a battery of verifications —that the macronutrient ranges allow a hundred per cent, that the fibre squares with its rule of fourteen grams per thousand kilocalories, that no reference value exceeds its own ceiling without a note explaining why — and if any of them fails, it refuses to write the book. One of those

checks caught a mistake of mine while I was writing this. That is why they are there.

HOW THIS BOOK IS READ

It can be read from beginning to end, and it is written for that. But it also works as a reference manual: the four acronyms in the chapter “The four acronyms nobody explains” are the key to everything else; the table of the nineteen is the reference you come back to every time a new tub appears; and the notebook at the end is the only thing to actually carry out. If you are only going to read three chapters, make it those three.

CHAPTER 2

THE FOUR ACRONYMS NOBODY EXPLAINS

Before the first figure you have to understand four acronyms, because they are the difference between reading a nutrition table and believing you are reading it. All four come from the same official framework, and the framework is more interesting than the figures.

THE REFERENCE FRAMEWORK, IN FOUR PIECES

EAR	Estimated Average Requirement. The amount that covers the need of HALF the population. It is the statistical starting figure, and it is nobody's target.
RDA	Recommended Dietary Allowance. It is worked out by adding two standard deviations to the EAR, so that it covers around 97 % of people. It is almost certainly more than you need.
AI	Adequate Intake. It is used when there is not enough data to calculate an EAR. It is a value observed in healthy populations, not a derived value: it carries less weight, and very few tables distinguish it from the RDA.
UL	Tolerable Upper Intake Level. The ceiling above which the risk of adverse effects starts to rise. It is not a toxic dose: it is where the known margin runs out.

THE CONCLUSION ALMOST NOBODY DRAWS

If the recommended value is set to cover ninety-seven per cent of the population, then most people are above their real need when they reach it. Falling a little below the RDA does not mean having a deficiency: it means you cannot tell, because a deficiency is defined against individual need and that figure appears in no table.

Hence “am I hitting a hundred per cent of the recommendation?” is a bad question, and the good one is “does my pattern of eating make it likely that something is missing?”. The second is answered by looking at the variety of the diet and, if needed, a blood test. The first is answered by an app that will give you a false number to two decimal places.

THE TECHNICAL MISTAKE ALMOST EVERY APP MAKES

The official documents themselves warn about this: the recommended value is for guiding an individual, and the average requirement is the one for assessing a group's intake. Using the recommended value to judge a population makes almost everyone look deficient. It is exactly the error that manufactures headlines and sells supplements, and it is advised against in writing in the very source those headlines cite.



AND ONE MORE PIECE, WHICH ARRIVED IN 2019

For sodium a new category was added: a chronic disease risk reduction value. It is neither a required intake nor a toxicity ceiling; it is the point from which lowering consumption is associated with less cardiovascular risk in the population. It will appear with figures in the sodium chapter, and it is worth knowing it exists because it is the first time this framework acknowledges that “enough” and “healthy” are not the same question.

CHAPTER 3

WHAT A DAY'S EXPENDITURE IS MADE OF

Before talking about how much to eat, it is worth knowing where the energy goes, because the real split looks nothing like what people imagine. Almost everyone believes exercise is the day's great drain. It is the smallest of the four.



THE PORTION EVERYONE COUNTS IS THE SMALLEST

COMPONENTS OF DAILY EXPENDITURE · TYPICAL RANGES

THE FOUR COMPONENTS OF DAILY EXPENDITURE Percentages of total expenditure, in typical ranges. Basal metabolism takes the largest share and can barely be changed at will; non-exercise activity is the component that varies most between people; structured exercise is, in most people, the smallest portion.

THE FOUR, ONE BY ONE

BASAL METABOLISM	60 to 70 % · what it costs to be alive at rest. The largest part of the day and the one least open to change at will
NON-EXERCISE ACTIVITY	15 to 30 % · walking, standing, gesturing, chores. It is the MOST variable component between people and the one that quietly collapses when you eat little
THERMIC EFFECT OF FOOD	8 to 12 % · what digesting and absorbing costs. Protein spends the most under this heading, by a wide margin
STRUCTURED EXERCISE	0 to 30 % · the part everybody counts and that in most people is the smallest of the four

THE UNCOMFORTABLE CONSEQUENCE

If structured exercise is at most a modest fraction of expenditure, then trying to offset food with exercise is a battle lost in advance by pure arithmetic. Half an hour of decent running is around four hundred kilocalories in a seventy-kilo person; a restaurant dinner can carry twice that in twenty minutes and without effort. This is not an argument against exercise: it is an argument against using it for what it is not good for.

And in the other direction there is better news. Non-exercise activity—walking, standing, taking the stairs, moving while you work—can differ twofold between two people of the same weight, and over the whole day it contributes considerably more than the gym session. It is the component you can influence most without taking time away from anything.



THE COMPONENT THAT QUIETLY COLLAPSES

When someone cuts back a lot on what they eat, and keeps it up, that component falls without the person deciding it or noticing: they move less, gesture less, take the stairs less, sit for longer. The practical consequence is that the real deficit ends up smaller than the planned one, and that accounts for much of the frustration of months three and four of any diet.

The antidote is not to squeeze harder. It is to deliberately protect everyday movement—a step target, taking the stairs, standing for phone calls—precisely in the phases when you are eating less. It is one of the most useful things you can do and it appears in no diet.

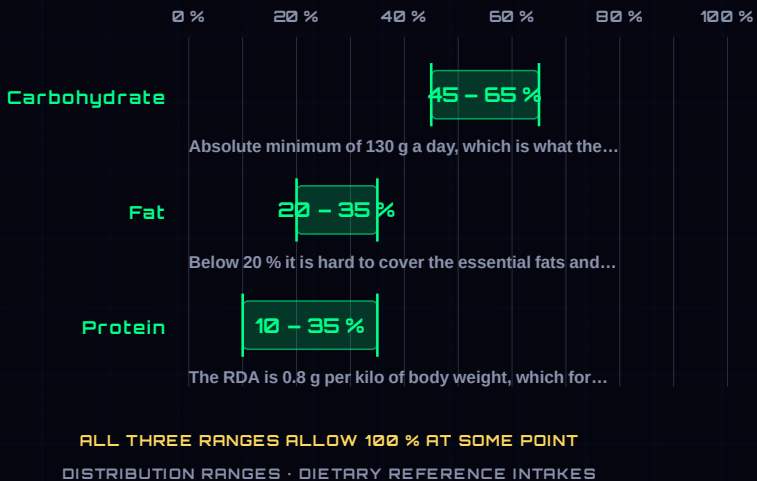
THE THERMIC EFFECT AND PROTEIN

Of the three macronutrients, the one that costs most energy to process is protein, by a clear margin. It turns nobody into a fat-burning machine—the magnitude is modest—but it explains part of why higher-protein diets come out somewhat better in weight-loss studies, along with satiety, which probably counts for more.

CHAPTER 4

HOW THE ENERGY IS DIVIDED UP

The first decision in any diet is not what to eat, but how the day's energy is divided between the three sources that provide it. And here the official answer is far wider than the public debate suggests.



THE THREE ACCEPTABLE RANGES, TO SCALE AGAINST A HUNDRED PER CENT Each bar is the range of the percentage of total energy that the reference values allow for that macronutrient. The three ranges overlap and leave ample room for very different diets: that is why a diet with thirty per cent protein and one with fifteen can both be correct.

WHAT THE RANGE SAYS

ACCEPTABLE DISTRIBUTION RANGES

CARBOHYDRATE	45 to 65 % of energy · absolute minimum of 130 g a day, which is what the brain consumes
FAT	20 to 35 % of energy · below 20 % it is hard to cover the essential fats and the fat-soluble vitamins
PROTEIN	10 to 35 % of energy · the RDA is 0.8 g per kilo of body weight, which for most people lands near 10 %

The important reading of this table is a negative one: within those margins there is no officially better split. The argument about whether more fat or more carbohydrate is better takes place entirely inside the permitted zone, and that is why we have spent thirty years failing to settle it.

What is outside the range is the interesting part. Going below the minimums has known consequences and leaves the zone where safety is established. The three minimums, with their reasons, appear in the next three chapters.

HOW TO TURN A PERCENTAGE INTO GRAMS

A gram of carbohydrate or protein provides four kilocalories; a gram of fat, nine. So thirty per cent of a 2,000 kcal diet is 600 kcal, and 600 divided by 9 is about 67 grams of fat. It works for any combination and it is all the arithmetic this book needs.



AND THE FOURTH MACRONUTRIENT, THAT NOBODY TALKS ABOUT

Alcohol provides seven kilocalories per gram and has no acceptable range, because it has no nutritional function: there is no recommended intake of alcohol and no body has ever set one. The dietary guidelines treat it only as a limit, and their explicit recommendation for anyone who does not drink is not to start. It is a fact that falls off almost every macronutrient table and that changes the accounting for anyone who drinks daily.

CHAPTER 5

PROTEIN: A FLOOR, NOT A TARGET

The official protein figure is the most cited and the worst understood in all of nutrition. It deserves a whole chapter because the misunderstanding runs in both directions at once: some believe it is an ambitious target and others that it is a safety ceiling, and it is neither.

THE FIGURE AND WHAT IT MEANS

RECOMMENDED VALUE	0.8 g per kilo of body weight per day
FOR 70 KG	56 g a day
ACCEPTABLE RANGE	10 to 35 % of energy
WHAT IT IS, EXACTLY	The RDA of 0.8 g/kg is the amount that prevents deficiency, not the amount that optimises muscle gain. It is a floor, not a target.

WHY IT IS A FLOOR

That value was set to answer one very specific question: how much protein a healthy adult needs in order not to lose nitrogen — that is, in order not to break down their own tissue. It is the threshold of non-deficiency. How much protein suits someone training to gain muscle is a different question, and the reference framework does not answer it because it never asked it.

The sports science literature has asked it, and its reviews converge on a considerably higher range, of the order of one and a half to two grams per kilo for someone training for strength with the intention of gaining tissue. I say very plainly that this range does NOT come from this book's official sources: it comes from scientific literature that is not in the public domain and that is only mentioned here. I say it because a book that mixes the two without warning makes you believe a government backs a figure it has not backed.

And I also say what the official sources do back, which is enough for almost everyone: the acceptable range goes up to thirty-five per cent of energy. In a 2,000 kilocalorie diet that is 175 grams of protein a day. Which is to say, the figures the gym works with fit comfortably inside the zone considered safe. You do not need to violate any official recommendation in order to eat plenty of protein, and that is the answer to half the arguments on the internet.



THE TWO THINGS THE TABLE DOES NOT SAY AND THAT DO MATTER

- Protein is not stored. Unlike fat and, to a small extent, carbohydrate, there is no reserve depot of amino acids. What is not used is oxidised or converted. Hence spreading it across the day makes more sense than concentrating it, even when the total is the same.
- Quality is measured by what is missing. A protein source is judged by the essential amino acid it is poorest in. Animal sources rarely fall

short; plant sources fall short predictably —cereals in lysine, pulses in methionine— and that is why combining them solves the problem without having to do it in the same meal.

- The weight used in the formula is body weight. In significant obesity that sum overestimates the need, because adipose tissue does not demand amino acids the way muscle does. It is one of the situations where the general formula stops working and you have to individualise with help.

WHAT MAY LEGALLY BE CLAIMED IN EUROPE

The European register of health claims authorises saying that protein contributes to the maintenance and growth of muscle mass, and also to the maintenance of bone. It does not authorise saying that a particular shake builds muscle by itself, or that it burns fat. The difference between those two sentences is exactly the difference between nutrition and advertising.

CHAPTER 6

CARBOHYDRATE: THE BRAIN'S HUNDRED AND THIRTY GRAMS

The carbohydrate minimum is the figure that best illustrates how a recommendation is built, because it does not come from a theory about diet: it comes from measuring what one organ consumes.

THE MINIMUM AND WHERE IT COMES FROM

RECOMMENDED VALUE	130 g a day, for any adult
ACCEPTABLE RANGE	45 to 65 % of energy
ORIGIN OF THE FIGURE	The brain's glucose consumption. That organ spends of the order of 100 to 140 grams a day and does not negotiate.
FIBRE APART	Fibre is a carbohydrate, but it has its own reference figure and its own chapter.

AND HERE IS THE PART ALMOST NOBODY TELLS YOU

The reference framework itself acknowledges that carbohydrate is not an essential nutrient in the strict sense, because the body can manufacture glucose from amino acids and from the glycerol of fats, and because the brain can cover a good part of its expenditure with ketone bodies when it is trained to. That is what makes very low-carbohydrate diets physiologically possible, and it is honest to say so in a book that gives a minimum figure.

What follows matters just as much: that something is possible does not make it recommendable by default, and the minimum of a hundred and thirty grams is set for the average adult in ordinary conditions, not as a verdict on a particular dietary strategy. Anyone who deliberately goes below it is outside the range where safety is established by consensus, and that is not a condemnation: it is information worth having before deciding, and a strong reason to do it with support.



ADDED SUGARS: THE GUIDELINES' ONE HARD LIMIT

The official dietary guidelines set an explicit cap on added sugars: less than ten per cent of daily energy. In a 2,000 kilocalorie diet that is 200 kilocalories, that is about fifty grams, which is roughly what a can and a half of soft drink contains.

It is worth noticing the word “added”. The sugar in a whole piece of fruit does not count towards that limit, and the sugar in a juice does count in practice, because it behaves like added sugar even though the label does not call it that. That distinction —the food matrix versus the molecule— is what explains why the guidelines talk about foods and not only about nutrients.

GLYCAEMIC INDEX: WHY IT DOES NOT APPEAR IN THE OFFICIAL TABLES

The glycaemic index measures the response to an isolated food, fasted and in a standardised amount, and those three conditions almost never hold in a real meal: fat, protein, fibre, processing and even the order in which you eat all alter the response. That is why the reference framework sets no glycaemic index targets. It is useful as a concept and poor as a shopping criterion.

CHAPTER 7

FAT: WHY THE MINIMUM IS TWENTY PER CENT

The fat range has the most interesting floor of the three, because it is not set by the fat itself. It is set by what the fat carries.

THE RANGE AND ITS TWO ENDS

ACCEPTABLE RANGE	20 to 35 % of energy
WHY NOT BELOW 20 %	Below that it is hard to cover the essential fatty acids and the fat-soluble vitamins, which need fat in the same meal in order to be absorbed.
SATURATED FAT	The guidelines limit it to less than 10 % of energy.
INDUSTRIAL TRANS FATS	No known safe intake. As low as possible.

THE TWO THAT ARE GENUINELY ESSENTIAL

Of all the fats, only two are strictly essential: linoleic acid and alpha-linolenic acid. The body cannot manufacture them and has to receive them. The rest, including the long-chain ones that get talked about so much, can be synthesised from those two, although the conversion is limited and there is legitimate debate about whether it is worth taking them directly.

Both have their own adequate intake in the official tables, and both are covered by very modest amounts of vegetable oils, nuts and seeds. It

is one of those things the supplement industry would rather you did not know.



THE FOUR FAT-SOLUBLE VITAMINS AND WHY THEY GO TOGETHER

Vitamins A, D, E and K share a property that makes them a group: they are absorbed with fat and stored in fatty tissue and in the liver. Two consequences follow, one good and one bad.

- The good one: you do not have to take them every day. The store covers long periods, which is why a vitamin D test reflects months of habit and not what you had for dinner yesterday.
- The bad one: excess is not eliminated in the urine either. These are the vitamins where careless supplementing does real harm, and three of the four have an established ceiling that can be reached with over-the-counter products.

THE INTERACTION THAT HAS TO BE SAID OUT LOUD

Vitamin K takes part in coagulation, and there lies the conflict with vitamin K antagonist anticoagulants. The solution is not to cut out leafy greens: it is to keep a STABLE intake and to make sure whoever adjusts the dose knows about it. What destabilises the treatment is the sudden change, not the amount.

CHAPTER 8

FIBRE AND WATER: THE TWO FIGURES THAT GET CALCULATED

These two references have something in common that makes them instructive: they are not numbers fallen from the sky, they are the result of a rule. And knowing the rule is worth more than memorising the number.

FIBRE: FOURTEEN GRAMS PER THOUSAND KILOCALORIES

THE RULE AND ITS TWO RESULTS

RULE	14 g of fibre per 1,000 kcal of the diet
MEN	38 g a day
WOMEN	25 g a day
WHY THEY DIFFER	The AI is set at 14 g per 1,000 kcal, and the two figures follow from the typical energy intake of each sex.

The practical consequence is direct: if you eat considerably less energy than the typical intake those figures were calculated on — because you are in a deficit, because you are small-framed, or because your expenditure is low— your fibre target is proportionally smaller, and you do not have to force thirty-eight grams. The rule scales; the number on the label does not.

In the other direction: someone eating 3,500 kilocalories because they train a lot needs close to fifty grams, and that does not happen by accident. It is one of the most common shortfalls in bulking diets built on shakes and white rice.

RAISING FIBRE ALL AT ONCE IS A CLASSIC MISTAKE

Doubling fibre in two days produces bloating, wind and, in some cases, more constipation rather than less. You raise it in steps of about five grams a week and you accompany it with water, because part of fibre's effect depends on there being liquid to form a gel with.



WATER: THE FIGURE ALMOST EVERYONE MISREADS

THE REFERENCE AND ITS SMALL PRINT

MEN	3.7 litres a day
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WOMEN	2.7 litres a day
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CAREFUL	This is TOTAL water: it includes the water in food and in every drink. Around 20 % usually comes from food, so water actually drunk falls well below that figure.
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This is **TOTAL** water: the water in drinks and the water in food. A piece of fruit is mostly water, and a soup, a yoghurt or a plate of vegetables contribute amounts that are anything but marginal. Food

accounts for around twenty per cent of the total, so the water you actually have to drink falls clearly below those figures. The eight daily glasses repeated everywhere come from no official document: they are a simplification that survived by repetition.

And a warning in the other direction, because it exists and gets forgotten: you can drink too much water. Exceeding the kidney's excretion capacity for hours on end dilutes the sodium in the blood, and that is a serious condition. It happens above all in long endurance events where people drink a great deal and drink only water. The sign that the balance is reasonable is reassuringly banal: clear urine several times a day and no thirst.

CHAPTER 9

THE NINETEEN, WITH BOTH THEIR COLUMNS

Here is the table almost nobody prints in full. Nineteen micronutrients with their reference value for men and for women and, on the same line, their ceiling. The right-hand column is the one missing everywhere else, and it is the one you need the day someone decides to buy themselves a tub.

	UNIT	MEN	WOMEN	CEILING · UL		
Vitamin A	µg RAE	900	700	3000		x3
Vitamin C	mg	90	75	2000		x22
Vitamin D	µg	15	15	100		x7
Vitamin E	mg	15	15	1000		x67
Vitamin K	µg	120	90	no ceiling		
Thiamin · B1	mg	1,2	1,1	no ceiling		
Riboflavin · B2	mg	1,3	1,1	no ceiling		
Niacin · B3	mg NE	16	14	35		x2
Vitamin B6	mg	1,3	1,3	100		x77
Folate	µg DFE	400	400	1000		x2
Vitamin B12	µg	2,4	2,4	no ceiling		
Calcium	mg	1000	1000	2500		x2
Iron	mg	8	18	45		x2
Magnesium	mg	420	320	350		x1
Zinc	mg	11	8	40		x4
Selenium	µg	55	55	400		x7
Iodine	µg	150	150	1100		x7
Potassium	mg	3400	2600	no ceiling		
Sodium	mg	1500	1500	no ceiling		

RDA ON THE LEFT · TOLERABLE LIMIT ON THE RIGHT · BAR = MARGIN

THE NINETEEN NUTRIENTS WITH THEIR REFERENCE AND THEIR CEILING Each row carries the recommended value and, where one exists, the tolerable upper limit. The rows with no ceiling are not the safest ones: they are the ones for which there were not enough data to set one, which is a different situation and not necessarily a better one.

HOW TO READ THE ROWS WITH NO CEILING

Nine of the nineteen have no established limit. It is tempting to read that as «this one can be taken carelessly», and it is the opposite of what

it means. A ceiling is set when there are enough data on adverse effects to fix a point; its absence indicates a lack of data, not an absence of risk. The official documents say so in these very words and it is still misread everywhere.



THE UNITS THAT TRIP YOU UP

Four nutrients are measured in units that build in a conversion, and if you do not know it, the arithmetic comes out wrong by a large factor. They are these:

UNITS THAT HIDE AN EQUIVALENCE

MG RAE · VITAMIN A	Retinol activity equivalents. One microgram of retinol is equivalent to twelve of dietary beta-carotene and to twenty-four of the other active carotenoids. Eating carrots is not the same as taking retinol, and the ceiling refers only to the latter.
MG DFE · FOLATE	Dietary folate equivalents. The folic acid in supplements and fortified foods is absorbed considerably better than natural folate, and the unit corrects for it. The ceiling refers only to the synthetic form.
MG NE · NIACIN	Niacin equivalents. The body makes niacin from tryptophan, at a rate of about sixty milligrams of the amino acid per milligram of the vitamin, and the unit includes it.
MG · VITAMIN E	Only alpha-tocopherol counts, and only certain stereoisomeric forms. A good part of the «vitamin E» on some labels does not count towards this figure.

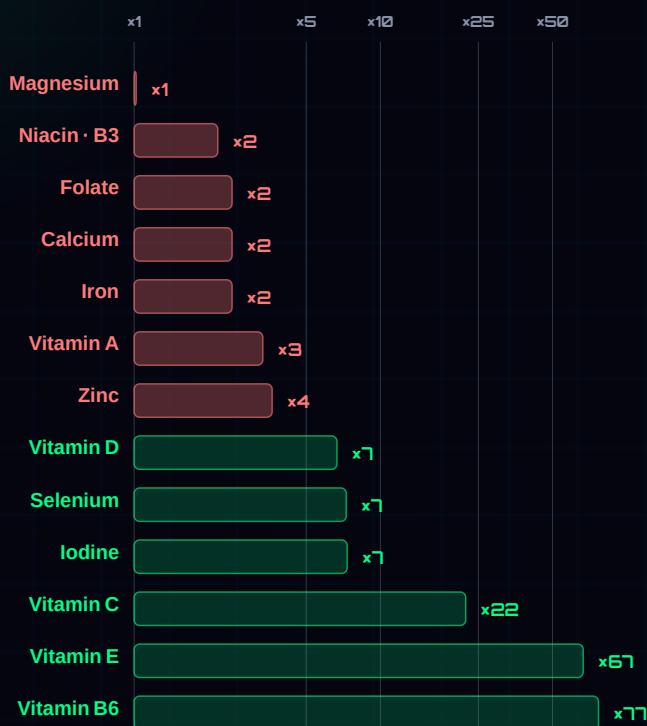
WHAT THE FRAMEWORK TAKES FOR GRANTED

These figures assume a mixed and varied diet, because bioavailability depends on what comes alongside. The documents themselves point to two specific cases: iron of plant origin is absorbed considerably worse, and a meat-free diet may require on the order of eighty per cent more iron; and a diet very rich in phytates —pulses and wholegrain cereals, which are otherwise excellent— can raise the need for zinc by as much as half. This is not an argument against plant-based diets: it is the small print you have to know in order to follow them well.

CHAPTER 10

THE MARGIN: WHICH ONES TOLERATE ERROR AND WHICH DO NOT

This is the plate that exists nowhere else, and the reason it was worth printing both columns. Dividing the ceiling by the recommended value gives a figure that says how many times over the recommendation can be taken before entering the zone with no known margin. Sorted by that figure, the nutrients separate in a way that comes as a surprise.



IN RED, A MARGIN OF LESS THAN SIX TIMES

LOGARITHMIC SCALE · CEILING DIVIDED BY REQUIREMENT

THE NUTRIENTS RANKED BY HOW NARROW THEIR MARGIN IS The ratio of the ceiling to the recommended value, on a logarithmic scale. The shorter the bar, the less room there is between what is recommended and what is best not exceeded. The bars marked in red are those with a margin below six times: there, blind supplementing can approach the ceiling without anyone noticing.

THE NARROW-MARGIN ONES, ONE BY ONE

These are the ones that demand you read the label and, in several cases, get a blood test first. They are not dangerous: they are the ones

that do not forgive the careless arithmetic of taking three products that all contain the same thing.

MAGNESIUM

REFERENCE	420 mg in men · 320 in women
CEILING (UL)	350 mg
WHAT YOU NEED TO KNOW	The only case where the UL looks lower than the RDA: the 350 mg UL refers ONLY to magnesium from supplements, not from food

It is the only case where the ceiling looks lower than the recommendation, and that apparent contradiction is one of the most useful things this table teaches: the ceiling refers ONLY to magnesium from supplements and medicines, not from food. You can eat magnesium with no worries; the laxative effect that sets the figure comes from concentrated salts, not from almonds.

IRON

REFERENCE	8 mg in men · 18 in women
CEILING (UL)	45 mg
WHAT YOU NEED TO KNOW	The 18 mg figure is for menstruating women. Supplementing without a blood test is a bad idea: the excess accumulates

The difference between the sexes is the largest in the whole table and it has an obvious cause. What is not obvious is the asymmetry of the error: falling short of iron can be corrected; going over it in a sustained way deposits iron in organs and there is no convenient physiological

route to get it out. There are also people with a genetic predisposition to absorb more than normal who do not know it. Of all over-the-counter supplements, this is the one it makes least sense to take without a blood test in front of you.

SELENIUM

REFERENCE	55 µg in men · 55 in women
CEILING (UL)	400 µg
WHAT YOU NEED TO KNOW	Narrow margin: excess is toxic and the commonest source of excess is Brazil nuts

The teaching case of this book. The Brazil nut is the most concentrated food source known, at something like seventy to ninety micrograms per nut, which means a generous handful can approach the daily ceiling with ordinary food and no supplement at all. It is no cause for alarm—one or two a day cover the requirement with room to spare—but a demonstration that «natural» and «no ceiling» are not synonyms.

VITAMIN B6

REFERENCE	1.3 mg in men · 1.3 in women
CEILING (UL)	100 mg
WHAT YOU NEED TO KNOW	Sustained excess produces neuropathy. It is one of the most frequently exceeded ULs, and unknowingly so

Here the margin looks wide, and in numbers it is; the problem is that the doses in commercial products are wide too. There are complexes carrying fifty or a hundred milligrams per capsule, that is, tens of times the reference, and the adverse effect described for sustained excess is a sensory neuropathy. It is the cleanest example of a ceiling crossed with no ill intent, by adding together two products that already contain it.

VITAMIN A

REFERENCE	900 µg RAE in men · 700 in women
CEILING (UL)	3,000 µg RAE
WHAT YOU NEED TO KNOW	The UL applies only to the preformed form (retinol), not to the carotenoids in vegetables

With the distinction in the unit already explained: the ceiling applies to preformed retinol, the kind in supplements, in liver and in some oils. The carotenoids in vegetables do not put it at risk. It is one of those times when two substances bearing the same name on the label do not behave the same way.



AND THE WIDE-MARGIN ONE THAT STILL NEEDS WATCHING

VITAMIN D

REFERENCE	15 µg in men · 15 in women
CEILING (UL)	100 µg
WHAT YOU NEED TO KNOW	15 µg is 600 IU. From age 71, 20 µg. The UL of 100 µg is 4,000 IU

I include it because it is the best-selling supplement and the most reasonable one on the list —cutaneous synthesis depends on latitude, season, clothing and age, and there are whole populations that do not get there through food— and because the margin invites complacency. It is worth keeping the two equivalences to hand: fifteen micrograms is 600 international units, and the ceiling of a hundred micrograms is 4,000. There are 10,000-unit products on the market.

THE RULE THAT SUMS UP THE CHAPTER

Eating brings nobody near the ceiling, apart from the curious case of Brazil nuts. Supplementing does, and supplementing at the same time with a multivitamin, a specific product and a fortified food is the usual way of getting there by accident. Before you add a tub, add up what you already take.

CHAPTER 11

SODIUM AND POTASSIUM: THE PAIR YOU READ TOGETHER

The last two in the table deserve a chapter of their own because they are the only pair that works like a scale, and because sodium is the nutrient for which the reference framework had to invent a new category.

SODIUM

REFERENCE	1,500 mg in men · 1,500 in women
CEILING (UL)	no established limit
WHAT YOU NEED TO KNOW	The AI is 1,500 mg. The chronic disease risk reduction intake (CDRR) sits at 2,300 mg, which is what about 6 g of salt carries

POTASSIUM

REFERENCE	3,400 mg in men · 2,600 in women
CEILING (UL)	no established limit
WHAT YOU NEED TO KNOW	It is an AI, revised in 2019. No UL for dietary intake

THE THREE SODIUM FIGURES

The adequate intake sits at 1,500 milligrams. The chronic disease risk reduction value sits at 2,300, which is the sodium in about six grams of salt, a little more than a level teaspoon. And actual

consumption in Western populations comfortably exceeds that second figure. The three together tell a story none of them tells alone: it is not that more sodium is needed, it is that it has to come down, and the realistic target is not the minimum but the second number.

And a clarification that is almost never made: most of the sodium in the Western diet does not come from the salt cellar. It comes from bread, cured meats, cheeses, tinned food, sauces and ready meals. Someone who takes away the salt cellar and goes on eating the same things barely moves the figure, and someone who cooks at home with salt is almost always better off than someone who does not use it but eats out of packets.



WHY POTASSIUM BELONGS IN THE SAME SENTENCE

Sodium's effect on blood pressure depends in part on how much potassium is eaten, and the dietary pattern the guidelines recommend works in both directions at once: more vegetables, fruit, pulses and tubers raise potassium and, by substitution, lower sodium, because they displace precisely the foods that carry it. It is a clear case of why recommendations are given as a pattern of eating and not as a list of molecules.

The potassium reference was revised downwards in 2019 relative to the previous one, and it still has no ceiling for dietary intake. For supplements it is another story, and there are medications —some

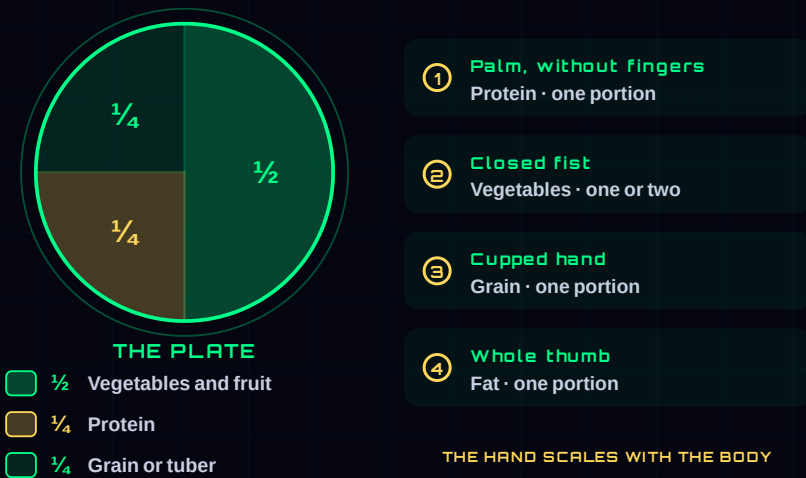
antihypertensives, some diuretics— that retain potassium. With renal insufficiency this paragraph reverses completely and you follow medical instructions, not a book.

THE CASE OF THE ATHLETE WHO SWEATED A LOT

In prolonged exercise in the heat, relevant amounts of sodium really are lost, and there the general recommendation to reduce it does not apply during the event. It is the only situation in this book where the population-level advice and the advice of the moment point in opposite directions, and it deserves saying so that nobody dehydrates through following a guideline written for another circumstance perfectly well.

THE PLATE AND THE HAND

Everything above is figures, and nobody eats figures. This chapter is the translation, and it uses two instruments you do not buy: the shape of the plate and the size of your hand.



NO SCALES · NO COUNTING · NO APP

THE DIVISION OF THE PLATE AND THE HAND AS A MEASURE Half the plate vegetables and fruit, a quarter protein and a quarter cereal or tuber reproduces, without counting anything, a division that falls inside the acceptable ranges. The four hand measures scale with the body: a big hand belongs to a big person, who needs more.

WHY THE HAND WORKS

The obvious objection to measuring with your hand is that it is not precise, and the answer is that neither are scales when they are used for two weeks and abandoned. The hand has two advantages no scales have: it is always available and it is proportional to whoever carries it. A ninety-kilo man has a bigger palm than a fifty-kilo woman and, without doing any arithmetic, his protein portion comes out larger in roughly the right proportion.

THE FOUR MEASURES

PALM, WITHOUT FINGERS	One protein portion: meat, fish, egg, tofu, cooked pulses
CLOSED FIST	One vegetable portion. One or two per meal, no arithmetic
CUPPED HAND	One portion of cereal, tuber or fruit
WHOLE THUMB	One dense-fat portion: oil, nuts, avocado, mature cheese

Two protein portions, two or three of vegetables, two of cereal and two of fat a day cover a reasonable pattern for a lot of people. Anyone training hard raises protein and cereal; anyone in a deficit lowers cereal and fat before protein and vegetables. It is not exact and does not claim to be: it claims to survive month three, which is where exact systems die.



THE SHOPPING LIST THIS BOOK PRODUCES

It is not a list of superfoods. It is the list of the categories that fill the gaps the table tends to show, ordered by what each one solves:

- Green leafy vegetables: folate, vitamin K, potassium, magnesium, fibre. The category that ticks the most boxes per unit of effort.
- Pulses: protein, fibre, non-haem iron, potassium, folate. With vitamin C in the same meal —a tomato, a pepper, a citrus fruit— iron absorption improves appreciably.
- Oily fish: the long-chain fatty acids, vitamin D and iodine. Twice a week is the standard advice in the guidelines.
- Egg: complete protein, choline, B vitamins, and fat-soluble vitamins in the yolk. Rehabilitated in the guidelines years ago; the blanket restriction once recommended no longer appears.
- Nuts and seeds: essential fats, magnesium, vitamin E, fibre. This is where the Brazil nut lives, so one or two and not a handful.
- Dairy or its fortified equivalent: calcium, protein, vitamin B12 and often vitamin D. If you replace it with a plant drink, check that it is fortified, because most unfortified ones supply none of this.
- Iodised salt: it is the route by which the population covers iodine. Without it and without fish or dairy, falling short is easy.

AND ONE MORE CATEGORY, FOR ANYONE WHO EATS NO ANIMAL PRODUCTS

Vitamin B12 is not present in plants in a usable form. It is not a question of planning better: it is a supplement or a fortified food, and it is not a matter of opinion. The rest of a well-run plant-based diet withstands any scrutiny; this particular point allows no improvisation, and sustained deficiency does neurological damage that is not always recovered.

CHAPTER 13

HOW TO READ A LABEL

The label on a packet is the only nutritional document most people have in front of them every day, and almost nobody knows how to read it. It has four known traps and a band of legal tolerance very few people suspect.

NUTRITION INFORMATION	
Per 100 g	ALWAYS COMPARABLE
Per serving	THE BRAND CHOOSES THE SERVING
Energy	
Fat · of which saturates	
Carbohydrate · of which sugars	
Fibre	
Protein	
Salt	

WHAT THE REGULATION SAYS

TO LIMIT

20 %

Calories, sugars, fat, sodium: at most one fifth ABOVE the declared figure

TO REACH

80 %

Vitamins, minerals, protein and fibre: at least four fifths of the declared figure

THE DECLARED VALUE IS A BAND

PUBLISHED LABELLING TOLERANCES

ANATOMY OF A LABEL AND ITS LEGAL TOLERANCES The per hundred grams column is the only one comparable between products. The serving is chosen by the manufacturer. And on the right, the tolerance: the declared value is not a number, it is a band.

THE FIRST TRAP: THE SERVING

The law requires values to be declared per hundred grams, and that is what makes two products comparable. Next to that column there is usually another one «per serving», and the serving is decided by whoever is selling. A thirty-gram serving of a breakfast cereal is roughly a quarter of what anyone actually pours, and every figure in that column is divided by four relative to what is really going to happen.

Rule: always compare per hundred grams, and then look at how much is really eaten. Never the other way round.

THE SECOND: THE ORDER OF THE INGREDIENTS

Ingredients are listed by descending weight, and that turns the list into a fairly honest X-ray. If sugar appears in the first three places, the product is mainly sugar however much the front of the pack talks about fibre. The industry's countermeasure is to split the sugar across several names —syrup, dextrose, juice concentrate, molasses— so that each weighs less and drops down the order. Adding them up in your head restores the real picture.

THE THIRD: THE WORDS THAT MEAN NOTHING

- «Natural» has no regulated definition that serves to judge anything.
- «Artisan», «home-made», «the way it used to be», «traditional recipe» are style, not composition.
- «No added sugars» does not amount to little sugar: a fruit juice qualifies and carries as much sugar as a soft drink.

- «Light» only requires a reduction relative to the reference product, which may still be extremely high in absolute terms.
- «Enriched with vitamins» usually marks a product that needed a reason to be in the trolley.



THE FOURTH, AND THE ONE NOBODY MENTIONS: THE TOLERANCE

Labelling rules allow a difference between what is declared and what is measured, and the figures are public. For what is best limited — calories, sugars, fats, sodium — the measured value must not exceed the declared one by more than a fifth. For what is best reached — vitamins, minerals, protein, fibre — the measured value must come to at least four fifths of what is declared.

For calories, sugars, total fat, saturated fat, cholesterol and sodium, the measured value must not exceed the declared one by more than a fifth. For vitamins, minerals, protein and fibre, the measured value must come to at least four fifths of the declared one. In other words: the label is a band, not a number.

Something practical follows: counting calories to two decimal places is fantasy arithmetic. A whole day of packaged food can carry an appreciable deviation from what was written down without anyone having broken any rule. That does not invalidate keeping a record; it invalidates false precision, and it argues for using trends over several weeks instead of balancing individual days.

THE THIRTY-SECOND READ

In practice three glances are enough for almost everything: the per hundred grams column, the first three ingredients and the salt figure. That is enough to tell a food from a product without having learnt any biochemistry.

EATING AROUND TRAINING

It is the question asked most often and the one that matters least, and it deserves a chapter precisely in order to put it in its place. The order of importance runs from top to bottom, and most people start at the last rung.

THE REAL ORDER OF IMPORTANCE

1 · THE DAY'S TOTAL	Total energy and protein. Almost everything is decided here, and with this unsorted the rest changes nothing.
2 · THE QUALITY OF THE PATTERN	That there are vegetables, fruit, pulses, lightly processed cereal and a decent protein source.
3 · THE SPREAD ACROSS THE DAY	Spreading protein over three or four meals instead of concentrating it. A modest and real effect.
4 · THE EXACT TIMING AROUND THE SESSION	The famous window. It is the smallest of the four rungs.

WHAT HAPPENED TO THE ANABOLIC WINDOW

For years it was taught that there was a sacred half hour after training in which protein had to go in or the session was lost. That urgency has not stood up well to scrutiny: if you have eaten in the preceding hours, the margin is hours and not minutes, and the day's total weighs far more.

Where timing does matter is in two specific and bounded situations: when you train fasted and the next meal is a long way off, and when there are two demanding sessions a few hours apart, which is an athlete's situation and not most people's.



BEFORE TRAINING

- A normal meal two or three hours beforehand works for almost everyone and requires nothing special.
- If you train shortly after eating, it helps to lower the fat and fibre in that meal: they delay gastric emptying and that is what produces the feeling of a stone.
- Training fasted is neither dangerous nor magic. It performs somewhat worse in long or very intense sessions and makes no difference in short ones. It is a preference, not a strategy.
- Coffee before training has a well-described effect on performance. Mind the hour: caffeine's half-life is around five hours, so an afternoon session can cost sleep that same night.

DURING

Below an hour of effort, water. Beyond that, and above all in the heat or in long events, it makes sense to supply carbohydrate and electrolytes: that is exactly the only authorised health claim for sports drinks, and the word doing all the work in that sentence is «prolonged».

AFTER

A normal meal with protein and carbohydrate, whenever it falls. If the session was very hard and the next one is tomorrow, there is no hurry at all. If there is another session in a few hours, then it does pay to eat soon and to prioritise carbohydrate, because the glycogen store takes time to refill and there the clock really is running.

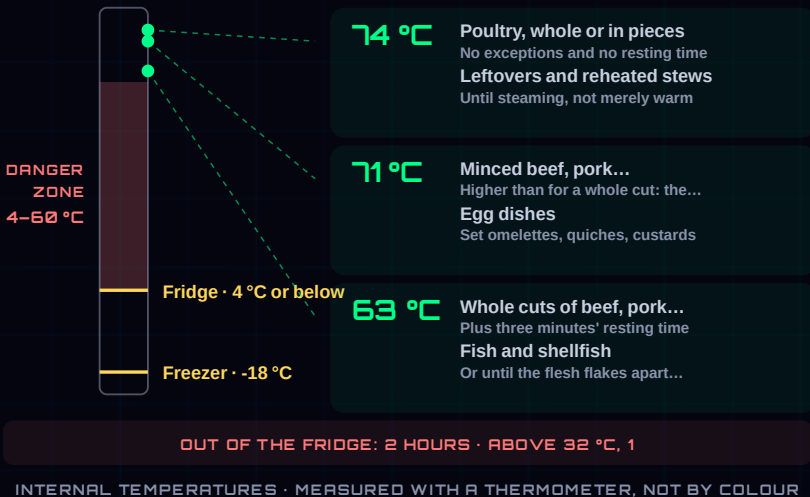
WATER AND THE WEIGHT AFTERWARDS

Losing a kilo during a long session is water, not fat, and it should be replaced over the following hours with drink and with food that contains salt. Weighing yourself right after training and celebrating the number is the commonest misreading in the gym.

CHAPTER 15

WHAT IS NOT COOKED PROPERLY DOES MORE HARM THAN WHAT IS EATEN BADLY

This chapter appears in no fitness book and ought to appear in all of them. An impeccable eating plan carried out with badly kept chicken produces more harm in two days than six months of perfect macros make up for. The figures that follow are public, from the United States food safety service, and they are simple.



INTERNAL TEMPERATURES · MEASURED WITH A THERMOMETER, NOT BY COLOUR

THE DANGER ZONE AND THE TEMPERATURES YOU HAVE TO EXCEED Between four and sixty degrees bacteria multiply rapidly. The safe internal temperatures are all above that band, and none of them is reached by colour: they are measured.

SAFE INTERNAL TEMPERATURES

POULTRY, WHOLE OR IN PIECES	74 °C · no exceptions and no resting time
LEFTOVERS AND REHEATED STEWS	74 °C · until it steams, not merely warm
MINCED BEEF, PORK, LAMB	71 °C · higher than for the whole cut: mincing spreads the surface through the inside
EGG DISHES	71 °C · set omelettes, quiches, custards
WHOLE CUTS OF BEEF, PORK, LAMB	63 °C · plus three minutes' resting time
FISH AND SHELLFISH	63 °C · or until the flesh flakes apart

WHY MINCE NEEDS A HIGHER TEMPERATURE THAN THE STEAK

It is the question that reveals whether someone has understood the matter. In a whole cut the contamination is on the surface, and the surface is precisely what receives the direct heat: that is why a steak that is rare inside can be safe. When the meat is minced, that surface is distributed throughout the mass, so the inside needs to reach the temperature by itself. An underdone burger is not the same thing as an underdone rib steak, and the difference is not one of taste.



THE FOUR STEPS

CLEAN	Hands for twenty seconds with soap, before and after touching anything raw. Boards and surfaces between one food and the next
SEPARATE	Raw away from ready to eat. A separate board for raw meat, and never the plate that carried the raw meat to the grill
COOK	With a thermometer and not by colour. The colour of meat does not indicate temperature: there is grey meat at 60 °C and pink meat at 75
COOL DOWN	Into the fridge within two hours. Large portions divided into shallow containers so that they come down fast

The step most often skipped is the fourth. Leaving a big pot to cool on the worktop all afternoon keeps it for hours inside the danger zone through sheer thermal inertia, even though the surface feels cold. The solution is to divide it into shallow containers and put them away promptly: they cool in a fraction of the time.

THE TIME MARGINS

AT ROOM TEMPERATURE	2 hours out of the fridge at most
ABOVE 32 °C	1 hour at most
FRIDGE	4 °C or less, checked with a thermometer and not with the control dial
FREEZER	-18 °C. Freezing does not kill bacteria: it stops them. On thawing they start multiplying again

THREE THINGS ALMOST EVERYONE GETS WRONG

- Thawing on the worktop. The surface enters the danger zone long before the centre thaws. You thaw in the fridge, in cold water changed every half hour, or in the microwave to cook immediately afterwards.
- Washing raw chicken under the tap. It does not remove bacteria and it splashes droplets more than half a metre around, contaminating the sink, the worktop and whatever is nearby. The heat of the pan does that job; the tap only spreads it about.
- Using the same plate to carry the raw meat to the grill and to bring it back cooked. It is the commonest route of cross-contamination in any domestic kitchen, and it is solved with a clean plate.

WHO HAS TO BE STRICTER

Pregnant women, older people, small children and people with compromised immunity carry an appreciably greater risk, and in that case some categories —raw-milk cheeses, raw egg, raw fish, uncured charcuterie, sprouted seeds— stop being a matter of preference. If you are in one of those groups, this gets asked about and not improvised.

CHAPTER 16

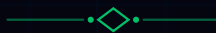
WHAT MAY BE CLAIMED, AND WHAT WAS APPLIED FOR AND REFUSED

This is the chapter that justifies half a collection. In Europe there is a public register recording every health claim that has been applied for on behalf of a nutrient, its scientific assessment and the verdict. The part that gets advertised is the authorised one. The interesting part is the other one.

HOW THE REGISTER WORKS

A company that wants to put on a packet that its product does something has to apply for authorisation of that sentence. A scientific panel assesses whether a sufficiently established cause-and-effect relationship exists between the nutrient and the effect, and issues an opinion. It is then authorised or not. The whole file stays public, rejections included.

That turns the register into something that exists in no other field: an official list of claims the industry wanted to make and could not sustain. It is not restricted material; it is public material almost nobody consults, and that is where the difference lies.



SOME AUTHORISED ONES, WITH THEIR SMALL PRINT

- Protein contributes to the growth and maintenance of muscle mass. Authorised, and that is why you see it on every shake container.
- Creatine increases physical performance in successive bursts of short-term, high-intensity exercise. Authorised with a specific condition of use, of the order of three grams a day. It is one of the very few ergogenic aids with an authorised claim, and it is worth knowing that the authorised sentence talks about short intense bursts, not about endurance or body composition.
- Carbohydrate-electrolyte solutions contribute to the maintenance of endurance performance during prolonged endurance exercise. Authorised; it is the legal basis of sports drinks, and the word «prolonged» does all the work.
- Vitamin D contributes to the maintenance of bones and to normal muscle function. Authorised. Note the word «normal»: the claim covers the maintenance of a function, not an improvement in performance.

AND SOME THAT WERE APPLIED FOR AND NOT AUTHORISED

- The glucosamine claims about cartilage and joints. Assessed and not authorised for insufficiently established cause and effect. It goes on selling enormously well.
- The branched-chain amino acid claims about muscle mass and recovery. Assessed and not authorised.

- The L-carnitine claims about fat oxidation and weight control. Assessed and not authorised.
- The weight-loss claims attached to green tea extracts. Assessed and not authorised.
- The term «antioxidant» as a generic claim. It is not an admissible claim on its own: the nutrient and the specific effect have to be stated, and only a few have that effect recognised.
- «Purifying», «detox», «alkalising» and the whole family. No authorised claim of that nature exists, which means that when you read one on a packet you are reading something that has passed no assessment at all.

THE REGISTER'S MOST CURIOUS CASE: CAFFEINE

The scientific panel assessed several caffeine claims favourably: alertness, concentration and endurance performance. Even so, those claims were never authorised, on public health considerations unrelated to efficacy: there was concern that they would encourage greater consumption. It is the only example I know of a substance that passed the scientific part and failed the political one, and it explains why you will not see that sentence on a can even though the effect is described.



THE LABEL-READING RULE THAT COMES OUT OF THIS

- 1 Look for the exact sentence. If it says «contributes to the normal function of», it is an authorised claim and it describes maintenance, not improvement.
- 2 Be wary of sentences that talk about a product instead of a nutrient. Claims are authorised for nutrients; no packet is authorised to say that it, specifically, does anything.
- 3 Check the condition of use. Many authorised claims require a minimum amount per serving, and a product can carry the nutrient in a token quantity and still name it.
- 4 If the claim is on the front in big letters and appears nowhere in the regulated wording, it is marketing and not an assessed property.

CHAPTER 17

THE FOUR SITUATIONS IN WHICH THE TABLE CHANGES

Everything above is written for a healthy adult. There are at least four situations in which the figures shift, and in two of them they shift so much that applying the general table is a mistake.

1 · FROM SIXTY-FIVE ONWARDS

WHAT CHANGES

PROTEIN	The population floor of 0.8 g/kg falls short for holding back the loss of muscle mass that comes with age. The geriatric literature works with clearly higher figures. This does not come from the reference tables: it comes from a different literature, and I say so.
VITAMIN D	The reference rises from age 71, and synthesis in the skin declines with age.
VITAMIN B12	Absorption worsens because of lower gastric acidity, which is why at this age the fortified-food or supplement form is often recommended, since it is absorbed better than the form found in meat.
CALCIUM	The reference rises for women from 51 and for men from 71.
WATER	The sensation of thirst declines with age, so the internal warning stops being reliable and drinking has to go by the clock.

2 · A DIET WITH NO ANIMAL PRODUCTS

It is a diet that withstands any scrutiny if it is followed attentively, and it has exactly five points that demand attention and not opinion:

- Vitamin B12: supplement or fortified food. No exceptions and no debate. Sustained deficiency produces neurological damage that does not always reverse.
- Iron: the plant-derived kind is absorbed considerably worse, and the official sources themselves indicate a notably greater requirement. Vitamin C in the same meal helps appreciably.
- Zinc: the phytates in pulses and wholegrain cereals reduce its absorption, and the requirement can rise by as much as half. Soaking, fermenting and sprouting reduces the effect.
- Iodine: without fish or dairy, iodised salt is the route. Seaweeds are an unpredictable source: some carry enormous amounts and exceed the ceiling easily.
- Long-chain fats: conversion from the plant ones is limited. There is a non-animal alternative for anyone who wants to cover it.

3 · ANYONE WHO TRAINS SERIOUSLY

What changes above all is protein, already dealt with, and total energy, which is what almost nobody adjusts. Eating in order to train five hours a week is not the same as eating in order to train one, and the commonest sign that energy has fallen short is not hunger: it is persistent fatigue, poor sleep, stalled performance and, in women,

disruption of the cycle, which is a sign that deserves a consultation and not an adjustment found on the internet.

4 · PREGNANCY AND BREASTFEEDING

Here this book stops. The references change for several nutrients, for some of them substantially, there is one where the margin is critical, and there are food categories with specific recommendations on microbiological risk grounds. It is the only situation in the whole book where I give no figure at all: you follow professional supervision and that is that.

AND A FIFTH THAT IS NOT A SITUATION BUT A WARNING

Any kidney, liver, heart or digestive disease reorders this table, and in some cases reverses it: there are nutrients that go from recommended to restricted. If you have a diagnosis, the table that counts is the one given to you by whoever is treating you, not this one.

HOW TO CHECK IT YOURSELF

A book that says «according to the official bodies» and does not say which one or where cannot be verified, and therefore cannot be corrected. This chapter is the checking apparatus: eight sources, all free, all online, with what to look for in each one.

THE SOURCES OF THIS BOOK

DIETARY REFERENCE INTAKES	National Academies · distributed by the NIH Office of Dietary Supplements. RDA, AI and UL for vitamins and minerals.
PHYSICAL ACTIVITY GUIDELINES FOR AMERICANS, 2ND ED.	U.S. Department of Health and Human Services, 2018. Activity dose and the response curve.
DIETARY GUIDELINES FOR AMERICANS	USDA and HHS. Dietary patterns and limits for added sugars, saturated fat and sodium.
FOODDATA CENTRAL	USDA Agricultural Research Service. Food composition, searchable online at no cost.
FM 7-22 HOLISTIC HEALTH AND FITNESS	U.S. Army. Doctrine on physical training, nutrition, sleep and mental readiness, published in full.
THE NIOSH LIFTING EQUATION	National Institute for Occupational Safety and Health, CDC. Calculation of the recommended weight limit in manual handling.
RESEARCH ON UNLOADING AND BED REST	NASA Human Research Program. Effects of unloading on muscle, bone and the cardiovascular system.
REGISTER OF HEALTH CLAIMS	European Commission, with the scientific assessments of EFSA. What may be claimed about a nutrient, and what was applied for and refused.



WHERE EACH THING IS LOOKED UP

CONCRETE ROUTES

RDA, AI AND UL	The fact sheets of the Office of Dietary Supplements at the United States National Institutes of Health, at ods.od.nih.gov . There is one per nutrient, in a consumer version and a health professional version, and it is the second that carries the complete tables.
FOOD COMPOSITION	FoodData Central, from the Department of Agriculture, at fdc.nal.usda.gov . You search for a food and it gives its complete profile, nutrient by nutrient.
DIETARY PATTERNS AND LIMITS	The Dietary Guidelines for Americans, at health.gov . The whole document downloads as a PDF.
DOSE OF PHYSICAL ACTIVITY	The Physical Activity Guidelines for Americans, second edition, at the same site. That is where the weekly dose of activity quoted in this book comes from.
TRAINING AND SLEEP DOCTRINE	FM 7-22, Holistic Health and Fitness, in the United States Army publications library. Published in full and downloadable.
BIOMECHANICS OF LOAD	The NIOSH lifting equation, at cdc.gov/niosh .
HEALTH CLAIMS IN EUROPE	The European Commission register. It can be filtered by nutrient and by status, and the status includes the non-authorised ones.

WHY UNITED STATES SOURCES IN A BOOK LIKE THIS ONE

For one purely legal reason and one practical one. The legal one: works of the United States federal government are not subject to copyright, which makes it possible to build on them a book that belongs entirely to whoever writes it. The practical one: they are the tables the rest of the world uses as a reference, the European agencies included, which publish their own with very close values. Where Europe differs in a relevant way, this book says so.



AND HOW THIS BOOK CHECKS ITSELF

The figures on these pages were not typed into the text. They live in a single data module that the typesetting program runs before writing a single line, and which refuses to go on if any of these checks fails:

- That the three macronutrient ranges admit a hundred per cent of the energy at some point.
- That the two fibre figures are consistent with the rule of fourteen grams per thousand kilocalories.
- That the nineteen micronutrient rows are complete and that no recommended value exceeds its own ceiling, apart from the case of magnesium, whose note explains why.
- That the equivalence between moderate and vigorous minutes adds up at both ends of the activity range.

- That every activity on the metabolic scale falls inside the intensity band it declares.

That last check caught a real error during the writing: I had classified an intensity as moderate when its value put it right on the threshold of vigorous. I corrected the datum, not the check. I mention it because a book that boasts of rigour ought to show the moment when rigour contradicted it.

THE TABLE NOTEBOOK

Four weeks, and not a diet. The notebook does not impose what to eat: it finds out what you eat, which is information almost nobody has about themselves and which no app supplies, because apps ask for what we believe we eat.

WEEK 1 · RECORD WITHOUT CHANGING ANYTHING

- 1 Write down what you eat, without weighing and without judging. Food and approximate size in hand measures. No figures.
- 2 Mark each meal with one of three letters: C if you cooked it, P if it came out of a packet, R if it was in a restaurant or similar.
- 3 At the end of the week count the letters. That split explains your intake of sodium, added sugars and fibre better than any calculation.
- 4 Change nothing this week. The measurement is contaminated if the intervention starts first.

WEEK 2 · THE THREE BOXES

- 1 Count how many meals in the week contained vegetables. Reference target: most of them.
- 2 Count how many contained a clear protein source the size of your palm.

- 3 Count how many days you had pulses, nuts or seeds.
- 4 Add ONE of the three where it is missing, whichever costs you least effort. One, not all three.

WEEK 3 · THE LIKELY GAPS

- 1 Go back over the list of categories in the plate chapter and mark the ones that never appear in your week.
- 2 For each gap, look in the table of nineteen to see which nutrients that category covered.
- 3 If the gap is oily fish, dairy or food of animal origin, that is where it makes sense to consider a blood test or a consultation, not a tub chosen online.
- 4 Write down the question you would ask a professional. One sentence. That sentence is the product of three weeks of notebook.

WEEK 4 · AUDITING THE MEDICINE CABINET

- 1 Gather every supplement and fortified food you take and write down, nutrient by nutrient, how much they add up to per day.
- 2 Compare each total with the ceiling column in the table of nineteen. Pay particular attention to vitamin B6, preformed vitamin A, iron, selenium, zinc and magnesium in salt form.
- 3 Any total that goes over the ceiling, or comes close, goes to the consultation. It is not adjusted by eye.
- 4 Anything you cannot justify with a specific reason—a documented deficiency, a diet that excludes a category, a stage of life—is a

candidate for disappearing. Most medicine cabinets lose half at this step and nobody notices a thing.

WHAT TO EXPECT FROM THE NOTEBOOK

It does not make you thinner. What it does is turn «I think I eat pretty well» into a description with data, and along the way save money on products that were doing nothing. If at the end of the four weeks you have a clear sentence to take to a professional and a smaller medicine cabinet, the notebook did its job.

WHAT YOU DO NOT ASK A TABLE

Last chapter, and the one that sets the limits. There are questions a reference table cannot answer, and pressing it to do so is the way popular nutrition turns into harm.

FIVE QUESTIONS THIS BOOK IS NO USE FOR

- «How much do I have to eat to lose X kilos in Y weeks?». This book is about nutrient sufficiency, not individual energy balance, and it distrusts on principle any figure promised with a deadline.
- «What do I take for my illness?». No figure here is established for clinical conditions, and several reverse in the presence of renal, hepatic or cardiac insufficiency.
- «Does this hold in pregnancy?». No. The references for pregnancy and breastfeeding are different ones, some of them considerably so, and there is at least one nutrient where the margin is critical. That stage is handled with supervision.
- «Is it any use for my child?». No. The whole table is for adults. Paediatric references change with age and with weight, and extrapolating by simple proportion is a mistake.
- «Am I deficient?». That is answered with a blood test and a clinical context, never with a food diary. No app has that information.

THREE SIGNS THAT YOU SHOULD CONSULT AND NOT READ

- Unintended weight loss, persistent tiredness, marked hair loss or any digestive change lasting weeks.
- Any plan that has you exclude a whole food group for more than a few weeks, whichever it is and however good the reason sounds.
- Any supplement you take without being able to explain in one sentence what it is for and how you will know whether it worked.



GLOSSARY

AI · ADEQUATE INTAKE	A reference value used when there are no data to calculate an average requirement. Less solid than an RDA.
AMDR · ACCEPTABLE RANGE	The range of the percentage of energy a macronutrient may supply.
BIOAVAILABILITY	The proportion of an ingested nutrient that actually becomes available to the body.
CDRR	Chronic disease risk reduction intake. A category introduced for sodium in 2019.
DFE · FOLATE EQUIVALENT	The unit that corrects for the different absorption of natural folate and synthetic folic acid.
EAR · AVERAGE REQUIREMENT	The amount that covers the need of half the population. It serves to assess groups, not individuals.
PHYTATE	A compound in wholegrain cereals and pulses that reduces the absorption of several minerals.
HAEM AND NON-HAEM IRON	The first, of animal origin, is absorbed considerably better; the second, from plants, depends on what comes alongside.
MET	Metabolic equivalent. A multiple of resting energy expenditure: one MET is roughly one kilocalorie per kilo of body weight per hour.
NE · NIACIN EQUIVALENT	The unit that includes the niacin made from tryptophan.
RAE · RETINOL EQUIVALENT	The unit that corrects for the different activity of retinol and of the carotenoids.

RDA · RECOMMENDED
INTAKE

The average requirement plus two standard deviations. It covers close to 97 % of the population.

UL · UPPER LIMIT

The ceiling above which the risk of adverse effects begins to rise.



And with that the book closes. It brings no diet at all, and that is exactly what it offers: the complete table instead of the advertisable half, the ceiling next to the recommendation, the units explained, the sources with a name and a route so that you can argue back at the book, and a warning repeated as often as necessary that no population figure is a prescription for a person.

If a year from now the only thing left of these pages is the habit of looking at both columns before buying a tub, and that of reading the label per hundred grams instead of per serving, the book will have done its job. Everything else is apparatus for arriving at those two habits.

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